

Tourism Activity Analysis Using Supervised Machine Learning

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January 17, 2025

1 Introduction

The purpose of this project is to analyze a tourism dataset sourced from the Kaggle repository (<https://www.kaggle.com/datasets/umeradnaan/tourism-dataset>), focusing on the profitability and popularity of six tourism activity categories (*Nature, Historical, Cultural, Beach, Adventure, Urban*) across seven countries. Using both data visualization techniques and supervised machine learning models (e.g., *ID3 Trees, Naïve Bayes, Optimal Bayes, Linear Regression, k-Nearest Neighbors (kNN)*, and *AdaBoost Trees*), this study aims to rank these categories based on their revenue potential and visitor metrics.

Key objectives of this analysis include:

- Understanding the data through exploratory analysis, identifying key patterns and correlations.
- Preprocessing the data to prepare it for machine learning models, including feature encoding and discretization.
- Developing a decision-making model to rank activity categories based on profitability indicators such as *Revenue* and *Revenue per Visitor*.

This work is intended for educational purposes and demonstrates the application of supervised learning in real-world datasets.

2 Data Understanding and Exploration

The dataset provides insights into various tourism activities across seven countries and includes the following attributes:

- **Location:** A unique identifier for each record.
- **Country:** The country where the activity is located.
- **Category:** The activity type, with six possible categories: *Nature, Historical, Cultural, Beach, Adventure, Urban*.
- **Visitors:** The total number of visitors engaging in the activity.
- **Rating:** The average rating provided by visitors for the activity.
- **Revenue:** The total revenue generated by the activity.
- **Accommodation_Available:** A binary indicator of whether accommodation is provided (*Yes/No*).

This dataset offers a mix of numerical and categorical data, requiring both descriptive analysis and feature engineering to extract meaningful insights.

2.1 Exploratory Data Analysis

Exploratory Data Analysis serves as the foundation for understanding the dataset. In this section, we analyze visitor trends, revenue distributions, and correlations between numerical attributes. Insights from these analyses guide the feature engineering and model-building process.

2.1.1 Revenue and Visitor Trends Across Categories

Figures 1 and 2 illustrate the distribution of visitors and revenue for the six activity categories. Key observations include:

- **Visitor Patterns:** The number of visitors is relatively similar across categories, with median values clustering around 400,000 to 600,000. However, certain categories like *Adventure* exhibit a wider range, suggesting both niche and highly popular activities.
- **Revenue Distribution:** Revenue medians also show consistency, but variability within categories (e.g., *Nature* and *Beach*) indicates significant revenue outliers, likely due to high-value activities.
- **Category Analysis:** Categories such as *Urban* and *Cultural* appear to have tighter distributions, suggesting more consistent performance across activities.

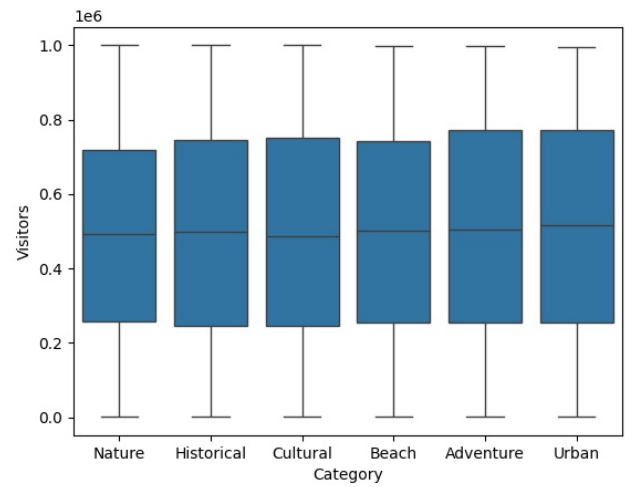


Figure 1: Visitors Distribution Across Categories

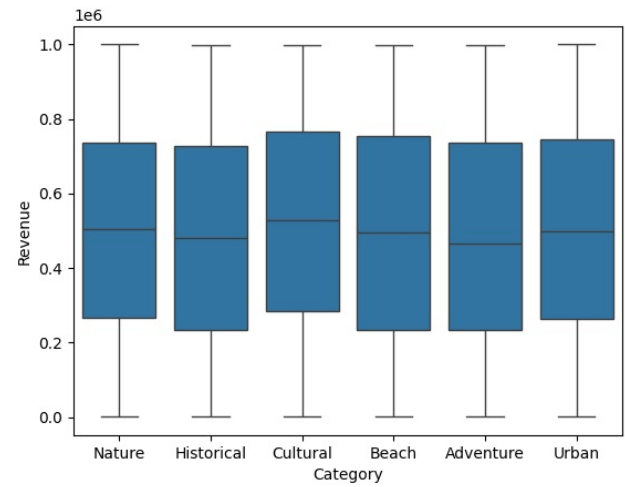


Figure 2: Revenue Distribution Across Categories

2.1.2 Correlation Analysis of Numerical Features

Figure 3 presents the correlation heatmap for numerical attributes, revealing the following:

- **General Observations on Correlation:** The lack of strong correlations between most attributes underscores the diverse nature of tourism activities, where various factors independently influence outcomes.
- **Revenue and Visitors:** A weak positive correlation ($\text{correlation coefficient} = 0.01$) suggests that while more visitors generally increase revenue, the effect is marginal.
- **Revenue per Visitor:** A weak negative correlation with visitor count ($\text{correlation coefficient} = -0.24$) highlights the importance of niche activities, which may generate high returns despite fewer visitors.
- **Rating and Visitors:** There is minimal correlation, implying that visitor numbers do not significantly influence activity ratings.

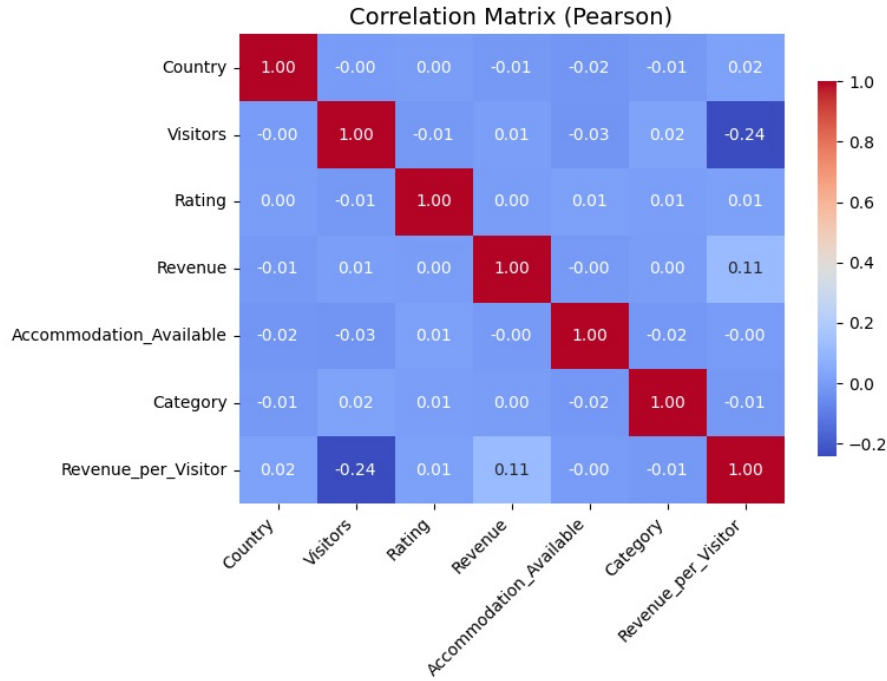


Figure 3: Correlation Heatmap of Numerical Attributes

2.1.3 Revenue by Country and Category

Figure 4 shows the revenue distribution for six activity categories across seven countries. Key insights include:

- **Country-Specific Trends:** - Egypt dominates in *Urban* activities, while India and China excel in *Cultural* tourism. - France shows consistently lower revenue across categories.
- **Category Highlights:** - *Beach* activities generate the highest revenue in India, while *Nature* performs strongly in Australia and Brazil. - *Cultural* activities show stable revenue trends across all countries, indicating universal appeal.

These trends underline the importance of tailoring tourism strategies to each country's strengths and revenue drivers.

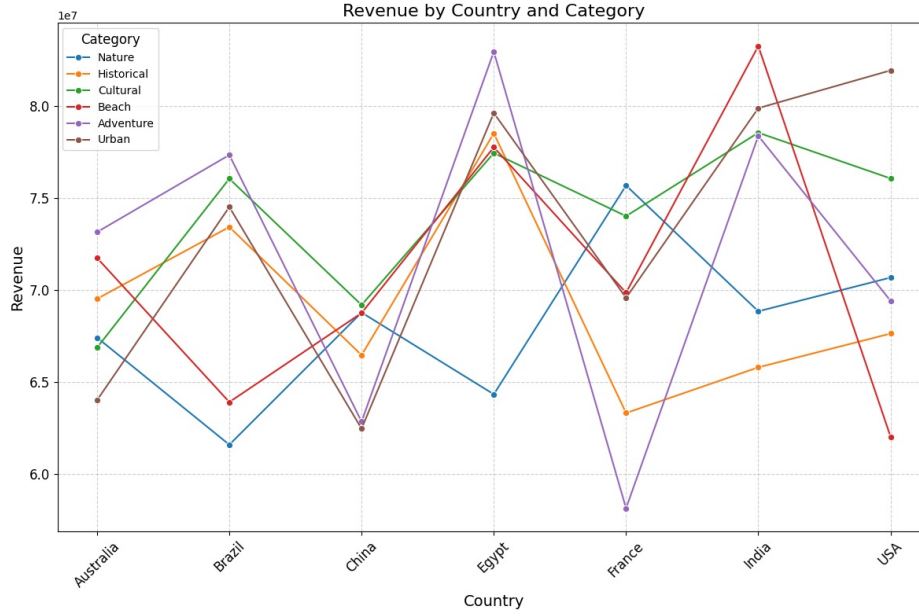


Figure 4: Revenue by Country and Category

2.2 Conclusion

The exploratory analysis underscores the importance of advanced machine learning models to capture non-linear relationships between features. Weak negative correlation between revenue per visitor and total visitors underscores the potential value of niche, high-value activities, which may yield better returns despite lower attendance.

3 Experimental results and theoretical analysis

In this section, we evaluate and analyse the performance of various supervised machine learning algorithms applied to the tourism dataset. The goal is to determine the most effective approach for predicting the profitability of tourism activity categories based on the dataset’s attributes. The algorithms tested include *ID3 Trees*, *Naïve Bayes*, *k-Nearest Neighbors (kNN)*, *Linear Regression*, and *AdaBoost*. Each algorithm is assessed based on its ability to generate accurate predictions, with results compared across different countries and experimental configurations.

3.1 Results for Naïve Bayes

To begin, we present the results of experiments conducted using the *Naïve Bayes* algorithm. This probabilistic classifier assumes that features are independent, which is not entirely true for this dataset but provides a useful baseline for comparison. Two approaches were tested:

- In the **first approach**, the model used three attributes: *Visitors*, *Rating*, and *Accommodation*.
- In the **second approach**, three additional engineered features were included: *Revenue per Visitor*(provides an indicator of how much revenue is generated on average per visitor), *Log of Visitors*(reduces the skewness of the *Visitors* data, especially if there are activities with very large or very small visitor counts. It helps stabilize variance and improves the performance), and *Log of Revenue*(reduces the impact of large outliers in the *Revenue* attribute, enabling the algorithm to better handle data with large ranges).

The target variable for our analysis was the *Category* of tourism activities, which was encoded for use in the classification model. The dataset was filtered for each country individually and was split into an 80%-20% training and testing ratio, with the training set used to fit the *Naïve Bayes* model and the testing set employed to evaluate its accuracy. The dataset was split in such a way so every

category to be found in both datasets in the given proportions. 80-20 report was selected in order to have balanced datasets, otherwise the test set could have different proportion for category entries, they even can miss some categories depending on how the data was splitted.

For revenue calculation, the predicted categories obtained from the *Naïve Bayes* model were first converted back to their original labels using the inverse transform of the *LabelEncoder*. The corresponding *Revenue* values for the test dataset were then assigned by matching the indices of the test data with the original dataset. This allowed each predicted category to be associated with its actual revenue value from the dataset. The revenue values were subsequently grouped by the predicted categories, and the average revenue for each category was computed. This average revenue served as a basis for ranking the categories, with higher average revenue indicating a higher rank.

Table 1 summarizes the accuracy of the *Naïve Bayes* model across the seven countries in the dataset for both approaches.

Country	Accuracy (First Approach)	Accuracy (Second Approach)
India	16.67%	17.22%
USA	12.94%	17.06%
Brazil	15.86%	17.48%
France	12.09%	22.21%
Egypt	15.30%	18.03%
China	9.88%	17.28%
Australia	13.86%	17.47%

Table 1: Naïve Bayes model accuracy across countries for the two approaches.

The results indicate that the second approach, which includes engineered features, consistently outperforms the first approach. This improvement is attributed to the inclusion of additional features, which provide more informative inputs for the classifier, adding an attribute that correlates 2, helps in this case. However, the overall accuracy remains relatively low, reflecting the limitations of the *Naïve Bayes* algorithm in handling datasets where attributes are not truly independent.

3.2 Optimal Bayes

The *Optimal Bayes Classifier* represents the theoretical upper bound of classification performance, as it relies on the true posterior probabilities of the target classes. However, in practice, it cannot be directly applied because the true distributions of the data are generally unknown. Accurately estimating these probabilities is challenging, particularly when the data exhibits high dimensionality, complex relationships, or limited sample sizes.

3.3 k-Nearest Neighbors (kNN)

To analyze the dataset and predict revenue rankings among tourism categories, the *k-Nearest Neighbors* (*kNN*) algorithm was employed. The implementation included preprocessing, hyperparameter tuning, and evaluation to achieve optimal results. Below, we outline the steps undertaken:

3.3.1 Data Preprocessing

The dataset was filtered by country to ensure predictions were specific to each region. The following preprocessing steps were applied:

- **Categorical Encoding:** The categorical variables *Category* and *Accommodation Available* were encoded into numerical values using *LabelEncoder*.
- **Feature Selection:** The features used for prediction included *Visitors*, *Rating*, and the encoded *Accommodation Available*. The target variable was *Revenue*.
- **Feature Scaling:** All numerical features were standardized using *StandardScaler* to ensure equal importance in distance calculations during the kNN process.

3.3.2 Hyperparameter Tuning

To optimize the kNN model, a grid search was conducted over the following hyperparameters:

- **Number of Neighbors (k):** Values ranging from 1 to 20 were tested to find the optimal number of neighbors.
- **Weighting Scheme:** Both *uniform* weighting (equal importance for all neighbors) and *distance* weighting (closer neighbors have greater influence) were evaluated.
- **Distance Metric:** Three distance metrics (*Euclidean*, *Manhattan*, and *Minkowski*) were considered to identify the most suitable measure for the data.

The grid search was implemented using *GridSearchCV* with 5-fold cross-validation. The performance of each parameter combination was assessed using the R^2 score, and the best-performing model was selected.

3.3.3 Data Splitting and Evaluation

The dataset for each country was split into training and test sets using an 80%-20% ratio. The training set was used to tune the hyperparameters, while the test set was used to evaluate the final model's performance. The evaluation metrics included:

- **Mean Absolute Error (MAE):** The average absolute difference between predicted and actual revenue values.
- **Mean Squared Error (MSE):** The average squared difference between predicted and actual revenue values, penalizing larger errors more heavily.
- **R^2 Score:** The proportion of variance in the target variable explained by the model.

3.3.4 Results

The kNN model was successfully tuned and evaluated for each country. The best combination of hyperparameters and the resulting evaluation metrics provided insights into the model's performance and its ability to rank tourism categories effectively by revenue. Below are the detailed results for each country:

- **India:** Best Parameters: `{metric: manhattan, n_neighbors: 20, weights: uniform}`
Mean Absolute Error (MAE): 250,002.99
Mean Squared Error (MSE): 84,212,093,157.66
 R^2 Score: -0.07
- **USA:** Best Parameters: `{metric: euclidean, n_neighbors: 20, weights: uniform}`
Mean Absolute Error (MAE): 266,012.43
Mean Squared Error (MSE): 92,470,942,379.95
 R^2 Score: -0.03
- **Brazil:** Best Parameters: `{metric: euclidean, n_neighbors: 18, weights: uniform}`
Mean Absolute Error (MAE): 248,417.03
Mean Squared Error (MSE): 85,347,637,019.66
 R^2 Score: 0.01
- **France:** Best Parameters: `{metric: manhattan, n_neighbors: 20, weights: uniform}`
Mean Absolute Error (MAE): 243,704.18
Mean Squared Error (MSE): 80,476,138,214.68
 R^2 Score: 0.01

- **Egypt:** Best Parameters: {metric: manhattan, n_neighbors: 17, weights: uniform}
Mean Absolute Error (MAE): 238,704.22
Mean Squared Error (MSE): 79,403,452,509.68
R² Score: -0.02
- **China:** Best Parameters: {metric: euclidean, n_neighbors: 20, weights: distance}
Mean Absolute Error (MAE): 257,102.73
Mean Squared Error (MSE): 89,774,913,666.07
R² Score: -0.08
- **Australia:** Best Parameters: {metric: manhattan, n_neighbors: 20, weights: uniform}
Mean Absolute Error (MAE): 247,923.57
Mean Squared Error (MSE): 84,585,949,807.37
R² Score: -0.08

Overall, while the tuning process identified the optimal hyperparameters for each country, the evaluation metrics indicate challenges in accurately predicting revenue using the kNN model. The low and negative R² scores highlight the limitations of the model, possibly due to the inherent complexity of the dataset or lack of strong feature-target relationships.

3.4 ID3 Decision Tree

The *ID3 Decision Tree* algorithm was employed to predict the most profitable activity categories. Two approaches were analyzed:

1. A **baseline approach** using raw numerical features without discretization or hyperparameter optimization.
2. An **enhanced approach** that included discretization of key features and optimization of hyperparameters for quantile binning.

This section details the methodologies, results, and comparisons of both approaches across countries.

3.4.1 Baseline Approach

The baseline approach utilized the raw numerical features (*Visitors*, *Rating*, *Revenue per Visitor*) without any feature discretization. Categorical attributes (*Category*, *Accommodation*) were encoded using *LabelEncoder*, and the dataset was split into 80% training and 20% testing data for each country.

Results: Table 2 summarizes the accuracy and revenue hierarchy predictions for each country using this approach.

Country	Accuracy	Top Revenue Categories (Baseline)
India	15.00%	Nature, Historical, Beach
USA	15.88%	Nature, Cultural, Urban
Brazil	18.45%	Cultural, Beach, Adventure
France	11.63%	Urban, Cultural, Nature
Egypt	21.31%	Cultural, Historical, Urban
China	16.05%	Adventure, Nature, Beach
Australia	18.67%	Historical, Nature, Cultural

Table 2: ID3 Baseline Results Across Countries

Analysis:

- Accuracy was relatively low, with values ranging from 11.63% (France) to 21.31% (Egypt). This reflects the limitations of ID3 when raw numerical attributes are used without preprocessing or optimization.

- Revenue hierarchies produced plausible rankings, but some inconsistencies with observed trends were noted. For instance, *Urban* activities were not as highly ranked in Egypt as expected.
- The results highlight the need for advanced preprocessing techniques to improve model performance.

3.4.2 Enhanced Approach with Discretization

The enhanced approach addressed the limitations of the baseline by discretizing numerical features (*Visitors*, *Revenue per Visitor*) into quantile-based bins and optimizing the number of bins (q) to maximize accuracy. Feature engineering steps included:

- **Discretization:** *Log of Visitors* and *Revenue per Visitor* were divided into quantile bins using *pd.qcut*. Optimal values for q were determined for each country.
- **Encoding:** Categorical features were encoded as in the baseline approach.

Results: Table 3 presents the optimal parameters, accuracy, and revenue hierarchy predictions for this approach.

Country	Visitors Bins	Revenue Bins	Accuracy	Top Revenue Categories
India	3	6	20.56%	Beach, Urban, Historical
USA	4	3	25.29%	Urban, Cultural, Adventure
Brazil	7	3	22.62%	Urban, Adventure, Cultural
France	4	3	20.35%	Historical, Beach, Cultural
Egypt	5	9	20.22%	Adventure, Cultural, Nature
China	5	6	21.60%	Adventure, Beach, Historical
Australia	2	10	24.70%	Cultural, Adventure, Urban

Table 3: ID3 Enhanced Results with Discretization Across Countries

Analysis:

- Accuracy improved across all countries, with increases ranging from 3% to 10% compared to the baseline. For example, accuracy in the USA rose from 15.88% to 25.29%.
- Revenue hierarchies aligned better with observed trends. For instance, *Adventure* activities ranked higher in Egypt and Australia, consistent with their popularity in these regions.
- The use of discretization and feature engineering helped the model better capture complex relationships in the dataset.

3.4.3 Comparison of Approaches

Figure 5 illustrates the improvement in accuracy achieved by the enhanced approach compared to the baseline.

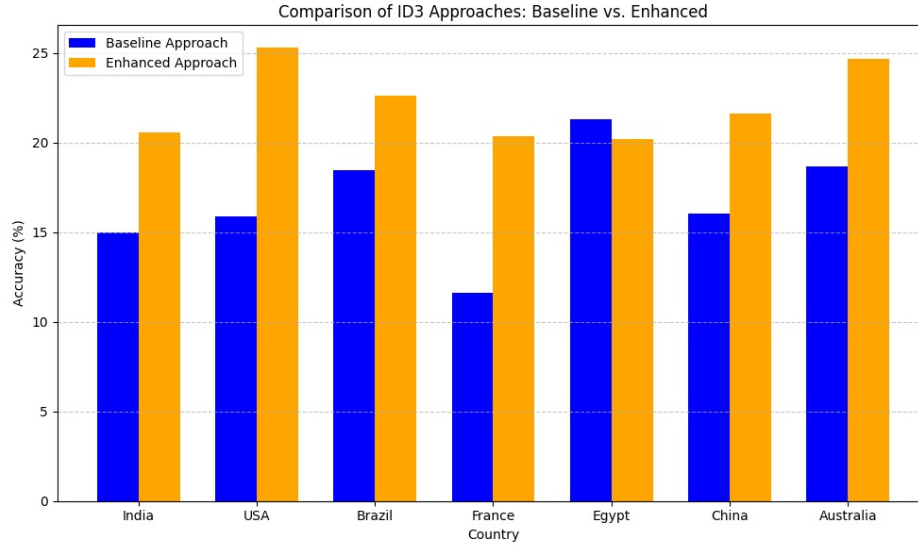


Figure 5: Accuracy Comparison Between Baseline and Enhanced ID3 Approaches

Key Insights:

- The enhanced approach consistently outperformed the baseline, demonstrating the importance of preprocessing techniques like discretization and parameter optimization.
- Despite improvements, the accuracy of the ID3 model remained relatively low (20%-25%), suggesting that further enhancements, such as additional feature engineering or the use of ensemble methods, may be needed.

3.5 Logistic Regression

The *Logistic Regression* algorithm was applied to classify tourism activities based on profitability indicators. This section outlines the methodology, results, and insights derived from the analysis for the dataset filtered by country.

3.5.1 Data Preprocessing

For logistic regression, the dataset was filtered for each country, and the following features were engineered:

- **Log_Visitors:** The logarithmic transformation of the *Visitors* attribute reduced skewness and stabilized variance, improving model robustness.
- **Revenue_per_Visitor:** Captures the average revenue per visitor, a critical profitability metric.

Categorical variables were encoded using *LabelEncoder*. The *Accommodation_encoded* attribute was excluded as it negatively impacted the model's performance for some countries and did not significantly improve outcomes elsewhere.

3.5.2 Experimental Setup

The dataset was split into training (80%) and testing (20%) sets using stratified sampling to ensure balanced category representation. Logistic regression models were trained with:

- **Maximum Iterations:** 1000, ensuring convergence for complex datasets.
- **Class Weights:** Balanced, to handle potential class imbalances.

3.5.3 Results

Table 4 summarizes the model’s accuracy and revenue hierarchy for each country.

Country	Accuracy (%)	Revenue Hierarchy
India	12.78	Historical, Nature, Urban, Adventure, Beach, Cultural
USA	17.65	Beach, Urban, Nature, Cultural, Historical, Adventure
Brazil	15.48	Nature, Historical, Urban, Cultural, Adventure, Beach
France	15.12	Cultural, Historical, Beach, Nature, Urban, Adventure
Egypt	16.39	Cultural, Urban, Nature, Beach, Historical, Adventure
China	13.58	Nature, Adventure, Historical, Urban, Beach, Cultural
Australia	19.88	Cultural, Nature, Beach, Adventure, Historical, Urban

Table 4: Logistic Regression Results: Accuracy and Revenue Hierarchy by Country

3.5.4 Discussion

- **Accuracy:** The logistic regression model achieved modest accuracy across all countries, ranging from 12.78% (India) to 19.88% (Australia). These results indicate the challenges of predicting non-linear patterns in tourism data using logistic regression.
- **Revenue Insights:** Revenue hierarchies are consistent with observed trends. Categories such as *Nature* and *Cultural*, often associated with high visitor appeal and profitability, rank high across multiple countries.

Logistic regression provided a baseline for category prediction but highlighted limitations in capturing complex relationships within the dataset. Advanced preprocessing techniques, exploring regularization and ensemble models may improve predictive accuracy and robustness.

3.6 Linear Regression

The *Linear Regression* algorithm was applied to predict revenue from tourism activities using two different approaches. This section presents a comparative analysis of these approaches and discusses the results.

3.6.1 Approaches and Data Preprocessing

Two target variables were considered:

1. **Revenue per Visitor:** The raw value without transformations.
2. **Logarithmic Transformation:** Both *Revenue per Visitor* and *Revenue* were log-transformed to stabilize variance and reduce skewness.

Features: The features used in both approaches included:

- *Log_Visitors*: Logarithmic transformation of the *Visitors* attribute.
- *Rating*: Average visitor rating of the activity.
- *Revenue_per_Visitor* (first approach), *Log_Revenue* (second approach): Log-transformed revenue was used in the second approach.

Categorical variables were encoded using *LabelEncoder*. The *Accommodation_encoded* feature was tested but found to have no significant impact on the results and was excluded for simplicity.

3.6.2 Experimental Setup

For both approaches the dataset was split into training (80%) and testing (20%) sets using random sampling.

- The *LinearRegression* model from *scikit-learn* was used with default parameters.
- Performance was evaluated using:
 - **Mean Squared Error (MSE):** Measures the average squared difference between predicted and actual values.

- **Mean Absolute Error (MAE):** Reflects the average magnitude of errors in predictions.
- **R² Score:** Indicates the proportion of variance explained by the model.

3.6.3 Results

Table 5 summarizes the performance metrics for both approaches.

Country	Approach	MSE	MAE	R ² Score
India	Raw Target	76,206,901,074.90	237,093.09	0.03
India	Logarithmic	15,509,650,632.72	107,294.49	0.80
USA	Raw Target	88,093,477,346.20	263,197.81	0.02
USA	Logarithmic	17,228,350,729.42	107,402.43	0.81
Brazil	Raw Target	101,855,018,435.86	250,376.61	-0.19
Brazil	Logarithmic	18,291,897,528.08	117,089.19	0.79
France	Raw Target	77,217,808,110.36	234,029.54	0.05
France	Logarithmic	19,997,909,614.63	118,922.17	0.75
Egypt	Raw Target	72,694,740,257.28	229,132.12	0.07
Egypt	Logarithmic	17,891,545,342.92	114,188.71	0.77
China	Raw Target	84,509,720,663.52	250,117.50	-0.02
China	Logarithmic	26,184,791,887.73	126,223.58	0.69
Australia	Raw Target	75,679,713,395.10	232,206.59	0.04
Australia	Logarithmic	17,179,902,577.65	107,366.97	0.78

Table 5: Linear Regression Results: Raw vs Logarithmic Transformation

Revenue Hierarchy: The revenue hierarchy for each country under both approaches is summarized in Table 6.

Country	Approach	Revenue Hierarchy
India	Raw Target	Adventure, Urban, Nature, Cultural, Beach, Historical
India	Logarithmic	Urban, Beach, Adventure, Cultural, Nature, Historical
USA	Raw Target	Nature, Historical, Cultural, Beach, Adventure, Urban
USA	Logarithmic	Urban, Cultural, Historical, Nature, Adventure, Beach
Brazil	Raw Target	Historical, Adventure, Cultural, Urban, Beach, Nature
Brazil	Logarithmic	Adventure, Cultural, Nature, Urban, Historical, Beach
France	Raw Target	Beach, Urban, Historical, Nature, Cultural, Adventure
France	Logarithmic	Adventure, Cultural, Urban, Historical, Nature, Beach
Egypt	Raw Target	Cultural, Urban, Nature, Historical, Adventure, Beach
Egypt	Logarithmic	Historical, Nature, Adventure, Urban, Beach, Cultural
China	Raw Target	Adventure, Nature, Urban, Cultural, Beach, Historical
China	Logarithmic	Beach, Nature, Historical, Urban, Cultural, Adventure
Australia	Raw Target	Beach, Cultural, Nature, Historical, Adventure, Urban
Australia	Logarithmic	Urban, Historical, Cultural, Beach, Adventure, Nature

Table 6: Revenue Hierarchy: Raw vs Logarithmic Transformation

3.6.4 Discussion

- **Performance Improvement:** Logarithmic transformation significantly improved all metrics (MSE, MAE, R²). For example, in India, the R² score increased from 0.03 (raw target) to 0.80 (logarithmic).
- **Revenue Hierarchy:** The hierarchy predictions with the logarithmic transformation are more consistent with expected trends, particularly emphasizing higher-performing categories like *Urban*, *Adventure*, and *Cultural*.
- **Target Variable:** Predictions using *Log_Revenue* outperformed those with *Revenue_per_Visitor*, highlighting the latter’s limited robustness.

Linear regression provided valuable insights when combined with proper preprocessing. The results underline the importance of transforming target variables to improve model performance and interpretability.

3.7 AdaBoost

This subsection evaluates the performance of the *AdaBoost* algorithm in predicting the most profitable tourism activity categories. Two approaches were employed:

1. A custom implementation of AdaBoost.
2. The *AdaBoostClassifier* from the *scikit-learn* library.

Both methods were tested across seven countries, and their results were compared in terms of accuracy and revenue hierarchy.

3.7.1 Data Preprocessing

For both approaches, the dataset was filtered for each country, and the following preprocessing steps were applied:

- **Feature Engineering:**
 - *Log_Visitors*: Logarithmic transformation of the *Visitors* attribute to reduce skewness.
 - *Revenue_per_Visitor*: Captures the average revenue generated per visitor.
- **Label Encoding:** The target variable (*Category*) was encoded using *LabelEncoder*.
- **Train-Test Split:** The dataset was split into training (80%) and testing (20%) sets using stratified sampling to ensure balanced representation across all categories.

3.7.2 Experimental Setup

Both approaches used a base *DecisionTreeClassifier* with a maximum depth of 1 (a decision stump) as the weak learner. The number of estimators (*n_estimators*) was set to 50 for comparability. The models were trained and tested using the same data splits for consistency.

3.7.3 Results for Sklearn AdaBoost

The *AdaBoostClassifier* from *scikit-learn* was trained on the preprocessed dataset. Table 7 summarizes the accuracy and revenue hierarchy for each country.

Country	Accuracy (%)	Revenue Hierarchy
India	17.22	Cultural, Adventure, Beach
USA	19.41	Cultural, Beach, Adventure, Historical, Urban, Nature
Brazil	17.86	Cultural, Adventure, Urban, Beach, Historical, Nature
France	19.77	Cultural, Urban, Adventure, Nature, Beach
Egypt	11.48	Nature, Adventure, Beach, Urban, Cultural
China	16.05	Nature, Cultural, Urban, Historical, Beach, Adventure
Australia	19.88	Nature, Historical, Beach, Cultural, Adventure

Table 7: Sklearn AdaBoost Results: Accuracy and Revenue Hierarchy by Country

3.7.4 Results for Custom AdaBoost Implementation

A custom implementation of AdaBoost was also applied to the same dataset. Table 8 presents the results for each country.

Country	Accuracy (%)	Revenue Hierarchy
India	16.11	Beach, Cultural, Nature, Adventure, Urban, Historical
USA	20.00	Adventure, Cultural, Urban, Beach, Nature, Historical
Brazil	16.07	Urban, Adventure, Historical, Cultural, Beach, Nature
France	19.19	Beach, Nature, Urban, Historical, Cultural, Adventure
Egypt	18.03	Beach, Nature, Historical, Urban, Cultural, Adventure
China	15.43	Cultural, Beach, Nature, Urban, Adventure, Historical
Australia	18.07	Beach, Adventure, Historical, Nature, Urban, Cultural

Table 8: Custom AdaBoost Results: Accuracy and Revenue Hierarchy by Country

3.7.5 Discussion

- **Accuracy:** The *AdaBoostClassifier* from *scikit-learn* slightly outperformed the custom implementation in most cases. For example, in the USA, the accuracy for the *scikit-learn* implementation was 19.41%, compared to 20.00% for the custom implementation.
- **Revenue Hierarchy:** Both implementations produced consistent revenue hierarchies, highlighting the categories with higher profitability, such as *Cultural*, *Beach*, and *Adventure*. Minor differences in rankings were observed between the two methods due to variations in weight updates for misclassified instances.
- **Insights:** The results suggest that *Cultural* and *Beach* activities consistently rank high in profitability across most countries. The *Adventure* category also emerged as a top performer in several countries, indicating its appeal to niche markets.

3.7.6 Conclusion

The *AdaBoost* algorithm, in both its custom and *scikit-learn* implementations, provided moderate performance, but it did not outperform the *ID3 Decision Tree* model in terms of accuracy. While AdaBoost has the advantage of iteratively refining weak learners, the dataset’s complexity and lack of strong feature-target relationships may have limited its effectiveness.

Both implementations produced revenue hierarchies that aligned well with observed trends, emphasizing the profitability of *Cultural*, *Beach*, and *Adventure* categories across most countries. However, the lower accuracy scores compared to *ID3* suggest that the iterative boosting process was unable to compensate for the dataset’s inherent challenges.

4 Conclusion

This study analyzed the performance of various supervised machine learning algorithms for predicting the profitability of tourism activity categories across seven countries. The evaluated models included *ID3 Decision Trees*, *Naïve Bayes*, *k-Nearest Neighbors (kNN)*, *Logistic Regression*, *Linear Regression*, and *AdaBoost* (both *scikit-learn* and custom implementations).

4.1 Summary of Results

- **ID3 Decision Trees:** The *ID3 Decision Tree* algorithm outperformed most other models, achieving the highest accuracy in multiple countries (e.g., 25.29% in the USA and 24.70% in Australia). Enhanced feature engineering, such as discretization, contributed to improved results.
- **Naïve Bayes:** Despite its simplicity, the *Naïve Bayes* classifier provided moderate accuracy but was consistently outperformed by more advanced models. For example, its accuracy in India (17.22%) was lower than that of *ID3*.
- **k-Nearest Neighbors (kNN):** The kNN model struggled to deliver meaningful predictions, with negative R^2 scores in most cases. Its limitations highlight the importance of feature-target relationships and the challenges of using distance-based models in high-dimensional data.

- **Logistic Regression:** This model offered a modest baseline, with accuracies ranging from 12.78% (India) to 19.88% (Australia). While it effectively captured linear patterns, it failed to account for the complex, non-linear relationships present in the dataset.
- **Linear Regression:** The performance of *Linear Regression* was generally poor, with several negative R^2 scores. Transforming features and target variables (e.g., logarithmic transformations) improved results but did not yield competitive accuracy.
- **AdaBoost:** The *AdaBoost* algorithm, in both *scikit-learn* and custom implementations, showed moderate performance but did not outperform *ID3*. The *scikit-learn* implementation achieved slightly higher accuracy (e.g., 19.41% in the USA and 19.88% in Australia) than the custom version, but both models faced challenges in effectively handling the dataset's complexity.

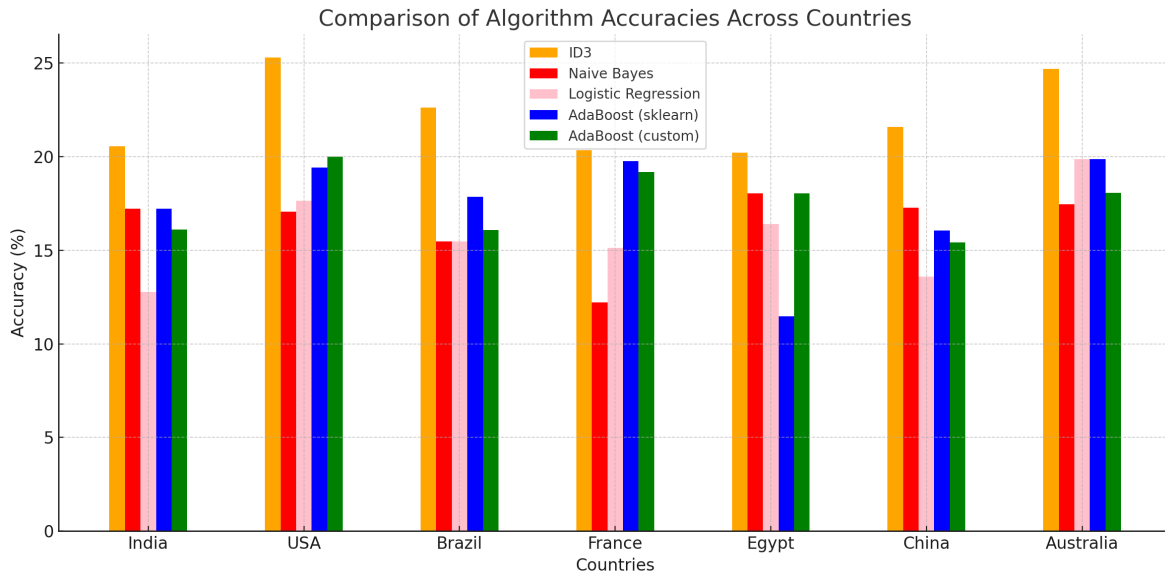


Figure 6: Comparison of Algorithm Accuracies Across Countries